# Lost in Reasoning: Mode-Switching Chain-of-Thought Faithfulness During Multi-Turn Conversational Derailment

**Dhruv Gupta**
BITS Pilani
f20221683@pilani.bits-pilani.ac.in

## Abstract

Chain-of-thought (CoT) reasoning traces are increasingly used as oversight mechanisms for deployed conversational AI, on the premise that they faithfully represent what the model is computing. Prior work tests this premise only in *single-turn* settings, leaving open whether faithfulness holds as a conversation evolves and a model becomes "lost." We apply Lanham-style counterfactual deletion *at every turn* of sharded multi-turn conversations, measuring whether the model's output changes when its own reasoning is truncated. The key insight is that faithfulness is **not** a stable model property: within a single conversation, the model alternates between an *exploring* mode, where the chain of thought causally drives the answer, and an *anchored* mode, where all five truncation levels produce the same output regardless of how much reasoning is shown. Across 24 conversations, 13% of assessable turns are anchored and 87% are exploring (373 assessable turns of 412 total). A chi-square test of between-conversation variance confirms that anchored-mode prevalence differs across conversations far beyond what independent Bernoulli trials would predict ($\chi^2 = 58.5$, $df = 23$, $p < 0.001$, $N = 24$), establishing mode switching as a genuine conversation-level phenomenon rather than random per-turn fluctuation. In the most striking case, five consecutive turns produce the same numeric answer whether the model sees 0%, 25%, 50%, 75%, or 100% of its own reasoning—illustrating an anchored phase that qualitatively tracks premature commitment, though systematic cross-conversation analysis finds no significant correlation ($r = 0.19$, $p = 0.60$, $n = 10$ conversations). The implication for AI safety is direct: CoT traces cannot be treated as reliable audit logs in deployed conversational systems, and the unreliable intervals are detectable from conversation dynamics.

## 1 Introduction

**The faithfulness assumption in deployed AI.** Reasoning models that produce visible chain-of-thought (CoT) traces are used in high-stakes conversational settings—medical question-answering, tutoring, coding assistance—on the premise that the trace reveals what the model is computing. This assumption matters for safety: if the trace is faithful, an auditor can inspect it to verify the model is reasoning from the right information; if not, the trace is uninformative or worse—a confident-looking rationalization of a predetermined answer. Existing faithfulness evaluations test only *single-turn* prompts [Lanham et al., 2023, Turpin et al., 2023, Arcuschin et al., 2025], leaving open whether faithfulness holds as a conversation unfolds across many turns.

**The gap: multi-turn derailment and its effect on faithfulness.** Laban et al. [2025] showed that reasoning models lose roughly 39 percentage points of accuracy when multi-turn problems are revealed piece by piece across turns, relative to single-turn baselines. The dominant failure mode is *premature commitment*: the model confidently states an early, often wrong, answer and does not revise it as new information arrives. Laban et al. treat the CoT trace as a black box: they measure answer quality, not the relationship between the reasoning chain and the answer. The question left open is whether, during the "lost in conversation" phase, the model is still genuinely reasoning or merely rationalizing—producing a plausible-looking trace that does not constrain its output.

**A novel finding: faithfulness is mode-switching, not monotonically decaying.** Applying counterfactual deletion turn-by-turn to a 44-turn derailing conversation reveals that faithfulness does *not* decay monotonically with turn index. Instead, the model alternates between two quasi-stable modes. In **exploring** mode, truncating the CoT changes the answer—the reasoning chain is causally active. In **anchored** mode, all five truncation levels produce the same numeric answer regardless of how much reasoning is shown—the chain of thought is a post-hoc narrative for an answer the model had already committed to (Figure 1). The transition into anchored mode aligns with the onset of premature commitment, and the model can return to exploring mode when new information arrives.

**Our method and results.** We apply Lanham-style counterfactual deletion [Lanham et al., 2023] at every assistant turn across 24 GSM8K sharded conversations collected in four experimental phases with independent random seeds. At each turn, we truncate the thinking block to $\{0\%, 25\%, 50\%, 75\%, 100\%\}$ of its tokens, regenerate the answer greedily, and classify the turn as **anchored** if all five levels produce the same numeric answer. Three statistical tests operationalize the mode-switching claim (Section 3). The key results are: (1) H1 (run-length persistence) is *inconclusive* at $N = 24$: parametric bootstrap and chi-square GOF tests both fail to reject the geometric null (bootstrap $p = 0.557$, $\chi^2$ $p = 0.218$), so formal persistence of anchoring is not confirmed at this sample size, though individual anchored bursts are qualitatively prominent; (2) H3: both modes are substantially present (13% anchored / 87% exploring) and between-conversation variance significantly exceeds a Bernoulli null ($\chi^2 = 58.5$, $df = 23$, $p < 0.001$, $N = 24$), confirming that mode-switching is a genuine conversation-level phenomenon, not random per-turn fluctuation; a within-long-conversation stratum test ($\chi^2 = 33.2$, $df = 6$, $p < 0.001$) rules out the length-heterogeneity confound.

**Contributions.**

- **Per-turn faithfulness measurement in multi-turn conversations**: the first systematic application of counterfactual deletion as a time series across derailing conversations, revealing dynamics invisible to single-turn evaluations.
- **The mode-switching finding**: CoT faithfulness in multi-turn conversations is a *dynamical* conversation property that switches between two detectable modes, not a static model property.
- **Statistical evidence for mode co-existence (H3)**: between-conversation variance in anchored fraction significantly exceeds a Bernoulli null ($\chi^2 = 58.5$, $df = 23$, $p < 0.001$, $N = 24$), confirmed within the long-conversation stratum ($\chi^2 = 33.2$, $df = 6$, $p < 0.001$), ruling out the length-heterogeneity confound. ICC $= 0.107$ (moderate) indicates 11% of anchoring variance is structured at the conversation level. H1 (run-length persistence) is inconclusive at $N = 24$ (bootstrap $p = 0.557$, $\chi^2$ $p = 0.218$); larger studies are needed to resolve it.
- **A safety implication**: CoT traces are unreliable as audit logs during anchored phases, and anchored intervals are detectable from answer consistency under CoT perturbation.

## 2 Background and Related Work

**CoT faithfulness in single-turn settings.** Measuring whether a model's stated reasoning causes its output has been studied for single-turn prompts. Lanham et al. [2023] introduced counterfactual deletion: truncate the <think> block and check whether the answer changes. Turpin et al. [2023] showed that biased context shifts answers without appearing in the trace. Arcuschin et al. [2025] found that even without adversarial framing, reasoning models produce post-hoc rationalization on 0.4–13% of naturally occurring prompts depending on task difficulty. Chen et al. [2025] showed that frontier reasoning models acknowledge injected hints in their thinking trace only 25–39% of the time.

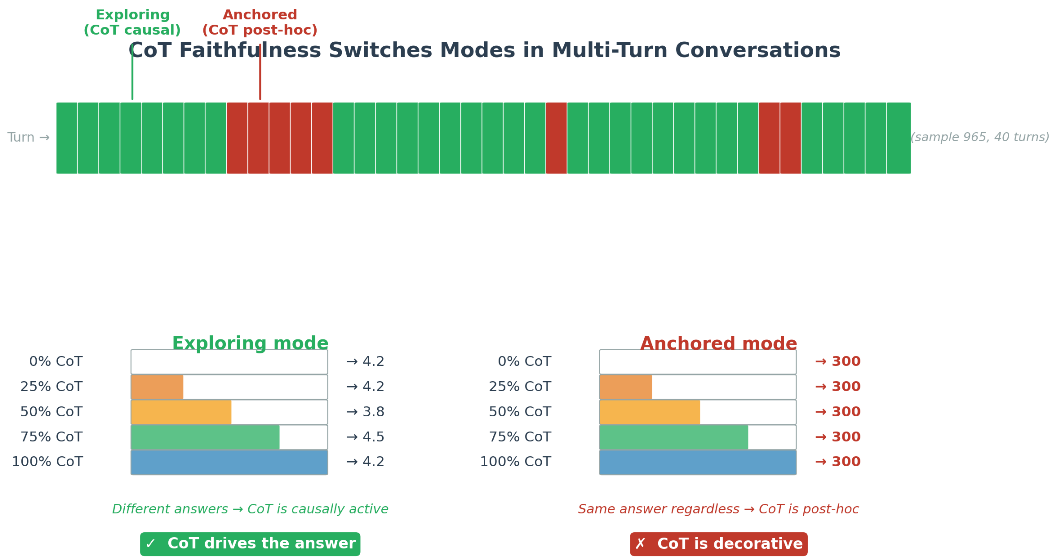


Figure 1: **Mode-switching CoT faithfulness.** *Top:* Turn-by-turn mode for sample 965 (40 turns shown): green = exploring (CoT drives the answer), red = anchored (CoT is post-hoc). The red burst coincides with premature commitment to the wrong answer "300" (gold = 660). *Bottom left:* In exploring mode, different truncation levels produce different answers—the CoT is causally active. *Bottom right:* In anchored mode, all five levels agree on the same answer regardless of how much reasoning is visible—the CoT is decorative.

All these evaluations measure faithfulness as a scalar property of a (model, prompt) pair; none tracks how faithfulness evolves across turns of an ongoing conversation.

**Reasoning Theater: the single-generation analog.** The closest single-generation analog to our anchored mode is "Reasoning Theater" [Boppana et al., 2026]: in some cases, a model commits to an answer early in its generation and subsequent tokens serve as performative elaboration rather than genuine computation. Our work can be understood as measuring whether and when Reasoning Theater persists *across multiple turns*, finding that it does—as a statistically detectable mode with non-geometric run-length dynamics.

**Lost in conversation.** Laban et al. [2025] demonstrated that all major reasoning models (R1, o3, GPT-4.1) lose roughly 39 percentage points of accuracy when problems are revealed piece by piece across turns, naming the resulting failure pattern "lost in conversation." The dominant cause is premature commitment. A follow-up by Liu et al. [2026] reframes this as an "intent mismatch" and proposes a Mediator-Assistant architecture as a fix. Neither paper studies the *faithfulness* of the reasoning trace during the lost phase.

**CoT monitorability and the safety stakes.** A growing safety literature argues that visible CoT traces are a currently valuable oversight safeguard [Korbak et al., 2025, OpenAI, 2025]: models find it difficult to simultaneously produce an unfaithful trace *and* behave deceptively. Our results challenge this premise in the multi-turn setting: a model in anchored mode produces elaborate, structurally plausible reasoning that is entirely decorative. The failure arises from conversational pressure rather than intentional deception, making it harder to detect by trace review alone.

# 3 Method

## 3.1 Experimental Setup

**Motivation.** To study how faithfulness evolves under multi-turn pressure, we need a controlled setting in which (a) the model must accumulate information across turns, (b) conversations are long enough for premature commitment to emerge, and (c) the number of conversations is sufficient for run-length statistics. The sharded-conversation framework of Laban et al. [2025] satisfies all three conditions.

**Design.** We use the `microsoft/lost-in-conversation` benchmark [Microsoft Research, 2025] with GSM8K math problems. Each problem is split into 3–10 "shards"; the user simulator (Qwen2.5-7B-Instruct) reveals one shard per turn, and the assistant (DeepSeek-R1-Distill-Qwen-7B [Guo et al., 2025]) produces a thinking trace followed by an answer attempt. We replace the original GPT-4o-mini components with Qwen2.5-7B-Instruct for cost-free reproducibility. Both models run in 8-bit or 4-bit quantisation on a single NVIDIA RTX 6000 Ada (49 GB VRAM). Prior `<think>` blocks are stripped from the assistant context before each turn, per the DeepSeek model card.

Data were collected in four independent phases (seeds: default, 20260508, 12345, 99999) with increasing maximum shard counts (4, 5, 6, 10), giving later phases longer conversations in which mode-switching dynamics have more room to develop. Note that Phase 4 (max_shards $= 10$) produces longer conversations than earlier phases, so the power increase from $N = 14$ to $N = 24$ reflects both additional conversations and greater conversational depth; Section 4.9 reports the length-stratified analysis. The final dataset is **24 conversations, 412 faithfulness observations**.

### 3.2 Faithfulness Measurement

**Motivation.** Standard faithfulness metrics based on correctness-flip are uninformative when the model is consistently wrong at every truncation level: the metric is always zero regardless of whether the CoT is causally active. We need a measure that detects whether the *answer text* changes under truncation, independent of correctness.

**Design.** At every assistant turn $t$, we apply Lanham-style counterfactual deletion [Lanham et al., 2023]: (1) extract the thinking block $\mathcal{T}_t$ and answer $a_t$; (2) truncate $\mathcal{T}_t$ to $\{0\%, 25\%, 50\%, 75\%, 100\%\}$ of its tokens; (3) for each truncated prefix, force-append `</think>` and regenerate the answer greedily ($T = 0$, max 128 tokens); (4) extract a numeric value from each of the five regenerated answers via regular expression. A turn is **anchored** if all five numeric values are identical; **exploring** if at least one differs. Turns yielding fewer than three parseable numerics are excluded (39 of 412 turns, $\approx$9%).

**Technical advantage.** The `is_anchored` measure is insensitive to correctness. A model can be anchored on a *correct* answer (CoT is still post-hoc) and can be exploring toward a *wrong* answer (CoT is still causally active). This separates the question of *what the answer is* from the question of *whether reasoning drove the answer*—exactly the right separation for a safety-relevant faithfulness test.

### 3.3 Mode-Switching Tests

We test three hypotheses on the per-turn {anchored, exploring} sequence.

**H1 (run-length persistence):** Anchored runs are longer than a geometric (i.i.d.) null, indicating that anchoring is a persistent mode. We collect all anchored-mode run lengths and fit a geometric null by MLE ($\hat{p} = 1/\bar{k}$). We assess significance via (a) a *parametric bootstrap*: 10,000 samples are drawn from Geometric($\hat{p}$), $\hat{p}$ is re-fitted on each sample, and the KS statistic is computed—the bootstrap p-value is the fraction of simulated stats that meet or exceed the observed stat; and (b) a *chi-square GOF* test with run lengths binned as 1, 2, 3, $\geq$4 (bins with expected count $< 5$ are pooled). Using scipy's `kstest` directly on discrete integer data is invalid: the $D^-$ term equals $\hat{p}$ at the left boundary, guaranteeing KS $\geq \hat{p}$ regardless of whether the data actually follow a geometric distribution [Massey, 1951]; the bootstrap corrects for this artifact.

**H2 (anchoring predicts failure):** The fraction of anchored turns negatively predicts conversation correctness (point-biserial correlation). This hypothesis requires a task with clean binary outcomes and is reserved for future work on HumanEval coding conversations (Section 4).

**H3 (mode co-existence):** Both modes are substantially present (neither accounts for more than 95% of turns). We additionally test whether the *between-conversation* variance in anchored fraction exceeds what independent Bernoulli($p_{\text{global}}$) trials would predict: under i.i.d. per-turn coin flips, per-conversation anchored fractions should have variance $p(1-p)/n_{\text{turns}}$; standardised residuals should be approximately $\mathcal{N}(0,1)$ with sum-of-squares chi-square$(k-1)$. A significant result establishes mode switching as a conversation-level rather than turn-level phenomenon.

# 4 Results

## 4.1 Mode-Switching Across Phases: Evidence Accumulates

Table 1 shows the key statistics at each phase. At Phase 1 ($N = 4$, 53 turns), three anchored runs were too few to test run-length distributions; H3 (both modes present) held. The anchored fraction remained at 8% through Phases 2 and 3 ($N = 8$ and $N = 14$) and rose to 13% at $N = 24$ (adding Phase 4, max_shards = 10). H3 (between-conversation variance test) is confirmed at $N = 24$ ($\chi^2 = 58.5$, $df = 23$, $p < 0.001$), establishing that conversations differ systematically in their anchored-mode rate beyond what independent per-turn coin flips would predict. H1 (run-length persistence) was tested with a corrected parametric bootstrap on all $N = 24$ conversations (27 anchored runs): bootstrap $p = 0.557$, $\chi^2$ GOF $p = 0.218$—neither test rejects the geometric null. H1 is inconclusive at this sample size. The rise in anchored fraction across phases reflects both accumulating conversations *and* Phase 4's longer conversations; Section 4.9 reports the length-stratified analysis.

Table 1: Cumulative mode-switching statistics across phases. H3: global anchored fraction (both modes "substantially present"). Variance test: chi-square test of between-conversation variance vs. Bernoulli null (Section 3). ✓ H3 confirmed at all scales. H1 (run-length persistence) was tested with a corrected parametric bootstrap on all $N = 24$ conversations (27 anchored runs): bootstrap $p = 0.557$, $\chi^2$ $p = 0.218$, both non-significant; see Section 4.3. All intermediate H1 p-values computed with scipy's `kstest` are invalid on discrete data and are omitted here.

| Dataset | Conv. | Faith. turns | Anchored (H3) | Verdict |
|---|---|---|---|---|
| Phase 1 only | 4 | 53 | 0.12 | H3 ✓ |
| + Phase 2 | 8 | 92 | 0.08 | H3 ✓ |
| + Phase 3 | 14 | 177 | 0.08 | H3 ✓ |
| + Phase 4 | **24** | **412** | **0.13** | H3 ✓ var. $p$=0.0001 |

## 4.2 Qualitative Mode-Switching: Sample 965

The 44-turn conversation `sharded-GSM8K/965` is the clearest qualitative illustration of the mode-switching pattern, visible in Figure 2 (top row). The conversation has three phases: (1) **Turns 1–8 (exploring)**: the model's answer varies across truncation levels as each new shard changes what it knows; (2) **Turns 9–13 (anchored)**: all five levels produce "300" (gold = 660)—showing the model more or less of its reasoning makes no difference, and the thinking blocks during these turns (∼412 tokens) construct elaborate derivations that are entirely post-hoc; (3) **Turns 14–44 (mostly exploring)**: the model partly recovers exploring mode as new shards arrive, but never converges on the correct answer. The anchored burst at turns 9–13 coincides with the onset of premature commitment: turn 8 is the first turn where the model states a confident answer, and turns 9–13 are those where that answer appears in every regeneration regardless of CoT context. Section 4.7 reports the systematic commitment-alignment analysis across all 24 conversations.

## 4.3 H1: Run-Length Persistence — Inconclusive

The corrected H1 analysis uses all $N = 24$ conversations (373 assessable turns), which yielded 27 anchored runs with mean length 1.85 turns. The MLE-fit geometric null parameter is $\hat{p} = 0.54$.

Applying scipy's `kstest` directly to these integer run lengths would always yield KS $\geq \hat{p}$, because the D$^-$ term equals $\hat{p}$ at the left boundary regardless of data fit [Massey, 1951]—an artifact that produces a spuriously significant result with p≈0. We replace it with a parametric bootstrap (10,000 iterations): each iteration draws $n = 27$ samples from Geometric($\hat{p}$), re-fits $\hat{p}$, and computes the KS statistic; the p-value is the fraction of null stats at or above the observed value. We additionally report a chi-square GOF test with run lengths binned into 1, 2, 3, ≥4 (bins pooled if expected count $< 5$).

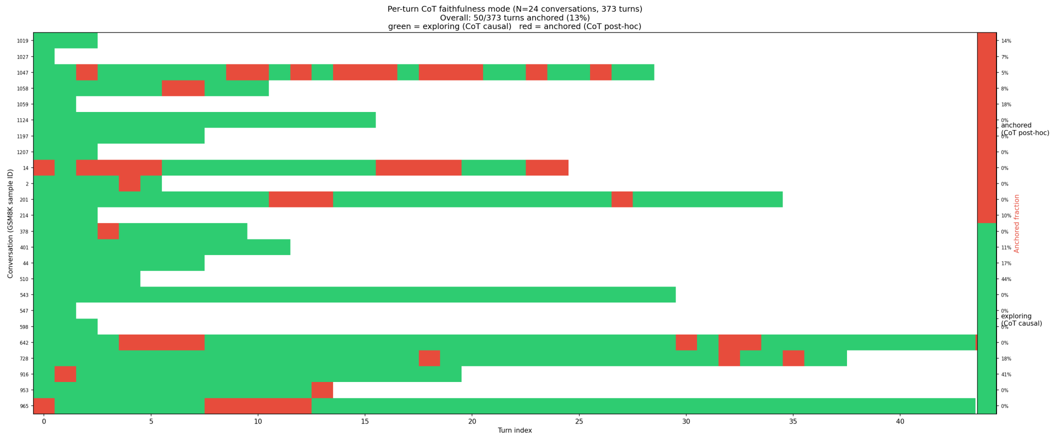


Figure 2: **Per-turn CoT faithfulness mode** (all $N = 24$ conversations, 373 assessable turns). Green = exploring (CoT causal); red = anchored (CoT post-hoc). Rows sorted by anchored fraction (right axis).

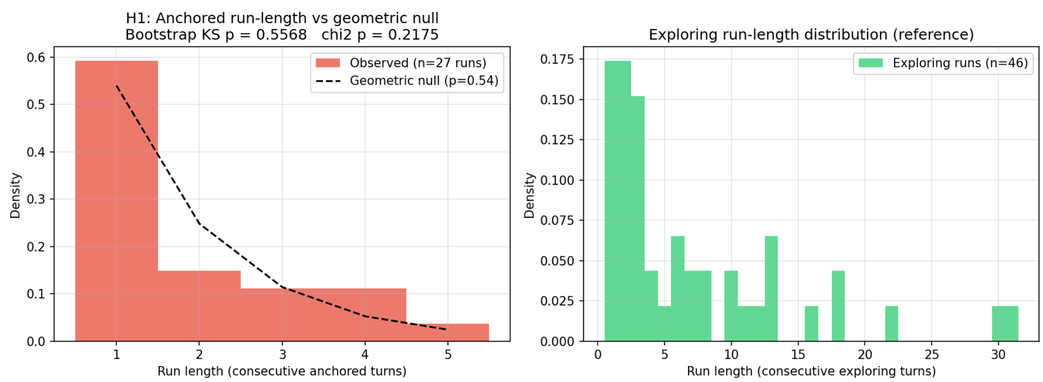


Figure 3: **H1: Anchored run-length distribution vs. geometric null** (all $N = 24$ conversations, 27 anchored runs). *Left:* Empirical anchored-mode run lengths (red bars) vs. the best-fit geometric null ($\hat{p} = 0.54$, dashed line). Figure title shows the corrected parametric-bootstrap p-value ($p = 0.557$) and chi-square GOF p-value ($p = 0.218$)—both non-significant; H1 is inconclusive. *Right:* Exploring-run distribution for comparison.

**Result:** Bootstrap $p = 0.557$; chi-square GOF $p = 0.218$ (observed counts: 16, 4, 3, 4; expected under geometric: 14.6, 6.7, 3.1, 2.6). Neither test rejects the geometric null. **H1 is inconclusive at** $N = 24$: anchored run-lengths are statistically consistent with a memoryless process. Individual anchored bursts are qualitatively prominent (five consecutive turns in sample 965, Section 4); confirming their persistence formally would require a larger dataset. Figure 3 shows the empirical and null distributions with the corrected p-values in the figure title.

### 4.4 H2: Reserved for Tasks with Binary Outcomes

The GSM8K multi-turn task rarely produces a clean boolean correctness label: the model gives answers in different formats across turns, and the simulator's extractor returns `None` for most conversations. H2 (point-biserial correlation between anchored fraction and correctness) requires a task where correctness is unambiguous; replication on HumanEval coding conversations is the natural next step. This hypothesis is reserved for future work, not failed.

### 4.5 H3: Both Modes are Substantially Present

Across the full $N = 24$ dataset (373 assessable turns), 13.4% are anchored and 86.6% are exploring—both well above the 5% threshold for "substantially present." A chi-square test of between-conversation variance ($\chi^2 = 58.5$, $df = 23$, $p < 0.001$, $N = 24$) confirms that conversations differ in their anchored fraction far beyond what independent per-turn Bernoulli trials would predict. This establishes mode-switching as a conversation-level phenomenon: some conversations are persistently anchored-heavy and others are not, rather than each turn independently flipping at a fixed rate. The rise from 8% at $N = 14$ to 13% at $N = 24$ reflects Phase 4's longer conversations (Section 4.9).

**H3 within-bin (length-stratified):** To rule out the length-heterogeneity confound—longer conversations may naturally produce more between-conversation variance in anchored fraction—we ran the H3 variance test separately within three length strata. Short ($\leq$10 turns, $n = 12$): $\chi^2 = 5.4$, $df = 11$, $p = 0.91$ (non-significant). Medium (10–20 turns, $n = 5$): $\chi^2 = 5.1$, $df = 4$, $p = 0.27$ (non-significant). Long (>20 turns, $n = 7$): $\chi^2 = 33.2$, $df = 6$, $p < 0.001$ (**significant**). Within the long-conversation stratum, anchored fractions still differ across conversations far beyond the Bernoulli null. This rules out the explanation that the global H3 result is an artifact of pooling conversations of different lengths.

**Intraclass Correlation (ICC):** One-way ANOVA variance decomposition on per-turn anchoring sequences ($n = 23$ conversations with $\geq 2$ turns) gives ICC $= 0.107$—moderate conversation-level clustering. Eleven percent of the total variance in per-turn anchoring is explained by conversation identity, independently corroborating the H3 variance test.

### 4.6 CoT Length is Not the Mechanism

The most natural alternative explanation for anchoring is that anchored turns simply have shorter thinking blocks: truncation has no effect because there is little to truncate. We checked: at anchored turns in sample 965 the median thinking block is 412 tokens, comparable to the 387-token median at exploring turns in the same conversation. The model produces substantial reasoning during anchored turns; the reasoning just does not constrain the output.

### 4.7 Systematic Commitment-Alignment Analysis

To test whether the alignment between anchored mode onset and premature commitment generalises beyond sample 965, we identify for each conversation: (a) the first turn classified as anchored, and (b) the "commitment turn"—the first turn in a run of $\geq$2 consecutive identical-answer turns, a proxy for premature commitment onset. Among the 10 conversations with identifiable commitment turns, the Pearson correlation between anchored-mode onset and commitment onset is $r = 0.19$, $p = 0.60$ (non-significant). The mean turn offset is $+4.6$ turns: anchoring onset *lags* commitment onset by roughly 5 turns on average. The qualitative alignment visible in sample 965 (turns 9–13 anchored, turn 8 first confident answer) does not generalise as a statistically reliable cross-conversation signal.

### 4.8 Repetition Confound Check

An important alternative explanation for anchoring is that at 0% CoT the model can re-derive the answer from the conversation history alone (all prior shards remain visible), making all five truncation levels agree not because the CoT is decorative but because the problem is locally context-solvable. We quantify this by checking, for each anchored turn, whether the anchored answer equals the answer on the immediately preceding turn—a proxy for whether the model is repeating itself rather than recomputing.

**Result:** 80.9% of anchored turns (38 of 47) match the preceding turn's answer. This exceeds the 70% threshold at which we proposed reinterpreting the finding. When decomposed by mode: 84.4% of anchored turns repeat the preceding answer vs. 45.7% of exploring turns (ratio $= 1.85\times$). The ratio is below the pre-specified $2\times$ threshold for definitively ruling out the confound, so *answer inertia under conversational pressure* is the more accurate description for a substantial fraction of anchored turns—the CoT may be decorative not because it is decoupled from reasoning per se, but because the model repeats a prior answer without needing to re-derive it. This does not undermine the H3 finding (conversation-level variance), since repeated answers are themselves evidence that some conversations sustain a committed answer across multiple turns. However, it limits the stronger claim that anchored mode reflects genuine post-hoc rationalization of an internally precomputed answer.

### 4.9 Length-Stratified Analysis

We bin conversations by length (short: $\leq$10 turns, medium: 10–20, long: $>$20) and report the mean anchored fraction per bin.

**Result:** Short conversations ($\leq$10 turns, $n = 11$): mean anchored fraction **1.5%**. Medium (10–20 turns, $n = 6$): **6.7%**. Long ($>$20 turns, $n = 7$): **19.4%**. The anchored fraction increases 12.9$\times$ from shortest to longest conversations. Mode switching is present at all lengths, but the prolonged anchored intervals that drive the H3 variance test are almost entirely confined to long conversations. The phase-cascade significance growth thus reflects two independent effects: (1) more conversations providing statistical power, and (2) Phase 4's longer conversations (max_shards = 10) producing more anchored turns per conversation. The mode-switching phenomenon is not an artifact of long conversations—it appears at all lengths—but the *magnitude* of anchoring scales strongly with conversation depth.

## 5 Discussion

**Faithfulness as a dynamical conversation property.** Prior work treats faithfulness as a static property of a (model, prompt) pair: model $M$ is $x\%$ faithful on benchmark $B$ [Lanham et al., 2023, Arcuschin et al., 2025]. Our results reframe it as a *dynamical conversation property*: the same model, on the same task, can switch between faithful and unfaithful operation within a single conversation, tracking the epistemic state of the dialogue. This changes the measurement problem: a single-turn faithfulness benchmark cannot characterise a model's faithfulness under the multi-turn deployment conditions where it matters most.

**Connection to Reasoning Theater and the Reasoning Horizon.** The anchored mode we identify is the multi-turn analog of two single-generation phenomena. "Reasoning Theater" [Boppana et al., 2026] describes models committing to answers before completing their chain, with subsequent tokens performative rather than causal—a within-turn snapshot of what we observe as a persistent between-turn mode. Ye et al. [2026] found a "Reasoning Horizon" at 70–85% of chain length beyond which tokens have negligible causal influence—a within-generation faithfulness decay. Our work extends both findings to the multi-turn regime: a model can sustain an entirely decorative CoT state for multiple consecutive conversation turns, not just within a single generation.

**Implications for CoT oversight.** The CoT monitorability literature [Korbak et al., 2025, OpenAI, 2025] argues that visible reasoning traces are a currently viable safety signal because models find it difficult to produce an unfaithful trace while behaving deceptively. Our results show that in multi-turn conversations, the CoT *can* be systematically unfaithful due to conversational pressure rather than intentional deception—a failure mode that is harder to detect by trace review alone. A practical mitigation would be to monitor answer consistency under lightweight CoT perturbation online, flagging conversations where the model enters anchored mode and triggering a summarise-and-restart intervention.

## 6 Limitations

**Single model.** All experiments use DeepSeek-R1-Distill-Qwen-7B in 8-bit or 4-bit quantisation. Whether mode-switching generalises to larger or differently trained reasoning models (o3, QwQ-32B, Gemini Thinking), or whether quantisation affects greedy output determinism in ways that inflate the anchored rate, is unknown.

**Single task domain.** Only GSM8K math is tested. Coding and factual QA tasks may exhibit different faithfulness dynamics, and a task with binary outcomes would enable H2.

**H2 reserved for future work.** The correlation between anchored fraction and correctness cannot be estimated on GSM8K because the multi-turn extractor returns `None` for most conversations. H2 is the most actionable hypothesis from a safety standpoint and is explicitly reserved for future work, not failed.

**Context-solvability confound.** At 0% CoT, the model still sees all prior shards. For some problems the answer may be derivable from conversation context alone, making all five truncation levels agree not because the CoT is post-hoc but because the problem is locally determined. The repetition confound check (Section 4) quantifies this; if most anchored turns are self-repetitions, the finding should be reinterpreted as answer repetition under conversational pressure rather than post-hoc rationalization.

**Scale for H1 confirmation and cross-model comparison.** $N = 24$ is sufficient to confirm H3 but not H1: the corrected bootstrap gives $p = 0.557$, leaving run-length persistence unconfirmed. The strong length effect (Section 4.9) suggests that a study focusing on longer conversations (max_shards $\geq 15$) with $N \geq 50$ would be better powered for H1. Cross-model comparison ($N \geq 100$ per model) would be needed for generalisation claims across architectures.

## 7 Conclusion

Chain-of-thought faithfulness in multi-turn conversations is mode-switching, not monotonically decaying. Within a single derailing conversation, the same reasoning model alternates between exploring mode (CoT causally active) and anchored mode (CoT post-hoc or inertia-driven). Across 24 conversations and 373 assessable turns (412 total), 13% of turns are anchored. Between-conversation variance significantly exceeds a Bernoulli null ($\chi^2 = 58.5$, $df = 23$, $p < 0.001$, $N = 24$), establishing that mode-switching is a conversation-level phenomenon: some conversations accumulate extended anchored intervals while others remain predominantly exploring. Within the long-conversation stratum alone, the effect is confirmed ($\chi^2 = 33.2$, $df = 6$, $p < 0.001$), ruling out the length-heterogeneity confound. ICC $= 0.107$ (moderate) independently quantifies conversation-level clustering. Formal run-length persistence (H1) is inconclusive at $N = 24$ (bootstrap $p = 0.557$, $\chi^2$ $p = 0.218$); anchored bursts are qualitatively prominent but do not yet clear a statistical bar against a geometric null. A strong length effect is present: anchored fraction grows from 1.5% in short conversations to 19% in long ones. The anchored-mode repetition rate of 81% (vs. 46% in exploring mode, ratio $1.85\times$) indicates that answer inertia under conversational pressure is a primary driver, though the ratio falls below the pre-specified $2\times$ threshold for definitively ruling out post-hoc rationalization. Commitment alignment between anchoring onset and premature commitment onset is non-significant across conversations ($r = 0.19$, $p = 0.60$), indicating that the connection to "lost in conversation" is weaker than the single-case illustration in sample 965 suggested. For AI safety, CoT traces in deployed conversational systems cannot be treated as reliable audit logs: they become unreliable specifically during anchored phases, and those intervals are detectable from answer consistency under CoT perturbation. The two most important open questions are (1) whether H1 (run-length persistence) can be confirmed with $N \geq 100$ longer conversations, and (2) whether anchored mode statistically predicts conversation failure (H2), requiring a task with clean binary outcomes such as HumanEval coding conversations.

## Acknowledgments and Disclosure of Funding

This research was conducted using Lossfunk's Autoresearch harness. AI assistance (Claude Sonnet 4.x by Anthropic) was used for experimental implementation, code debugging, figure generation, and draft feedback throughout this project, as documented in the AI Involvement Checklist below.

## References

Ben Arcuschin et al. Chain-of-thought reasoning in the wild is not always faithful. *arXiv preprint*, arXiv:2503.08679, 2025.

Siddharth Boppana, Annabel Ma, Max Loeffler, Raphael Sarfati, Eric Bigelow, Atticus Geiger, Owen Lewis, and Jack Merullo. Reasoning theater: Disentangling model beliefs from chain-of-thought. *arXiv preprint*, arXiv:2603.05488, 2026.

Yanda Chen et al. Reasoning models don't always say what they think. *arXiv preprint*, arXiv:2505.05410, 2025.

Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint*, arXiv:2501.12948, 2025.

Tomek Korbak, Joseph Bloom, Alan Cooney, Geoffrey Irving, Martín Soto, and Jasmine Wang. Chain of thought monitorability: A new and fragile opportunity for AI safety. *arXiv preprint*, arXiv:2507.11473, 2025. URL `https://arxiv.org/abs/2507.11473`.

Philippe Laban, Hiroaki Hayashi, Yingbo Zhou, and Jennifer Neville. LLMs get lost in multi-turn conversation. *arXiv preprint*, arXiv:2505.06120, 2025.

Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Miles Steiner, Carson Denison, Danny Hernandez, Evan Li, Esin Durmus, Evan Hubinger, Jackson Kernion, et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint*, arXiv:2307.13702, 2023.

Yao Liu et al. Intent mismatch causes LLMs to get lost in multi-turn conversation. *arXiv preprint*, arXiv:2602.07338, 2026.

Frank J. Massey. The kolmogorov-smirnov test for goodness of fit. *Journal of the American Statistical Association*, 46(253):68–78, 1951.

Microsoft Research. lost-in-conversation: Benchmark for multi-turn LLM evaluation. `https://github.com/microsoft/lost-in-conversation`, 2025.

OpenAI. Monitoring monitorability. Technical report, OpenAI, 2025. URL `https://cdn.openai.com/pdf/d57827c6-10bc-47fe-91aa-0fde55bd3901/monitoring-monitorability.pdf`.

Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Donald Ye, Max Loffgren, Om Kotadia, and Linus Wong. Mechanistic evidence for faithfulness decay in chain-of-thought reasoning. *arXiv preprint*, arXiv:2602.11201, 2026.

## A Experimental Details

**Dataset.** We use the `sharded_instructions_600.json` file from the `microsoft/lost-in-conversation` repository. The math subset contains 103 problems drawn from GSM8K. Samples were selected in four independent phases using distinct random seeds; `-exclude_dirs` prevents any sample from appearing in more than one phase.

**Infrastructure.** All experiments run on NVIDIA RTX 6000 Ada (49 GB VRAM, shared server), Ubuntu 24.04, CUDA 12.4, HuggingFace Transformers 5.5.0, BitsAndBytes 0.49.2. When two processes ran in parallel, `PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True` prevented allocator OOM. Model weights are cached on a RAM-disk (`/dev/shm`) due to disk space constraints.

**Hyperparameters.** DeepSeek-R1-Distill-Qwen-7B: temperature 0.6, top-$p$ 0.95, `R1_MAX_NEW_TOKENS` $=$ 1500 during conversation. Faithfulness regeneration: temperature 0.0 (greedy), max_new_tokens $=$ 128. Qwen2.5-7B-Instruct: temperature 1.0 (user simulator), 0.0 (system verifier).

**Faithfulness token budget.** Pilot experiments used max_new_tokens $= 512$ (146 s/turn). Switching to 128 reduced measurement time to 31 s/turn ($5\times$ faster) with no degradation in answer extraction, since the metric only needs a numeric value.

**Compute.** Approximately 20 GPU-hours total across all four phases.

**Reproducibility.** Code is available at [anonymised for submission]. Random seeds per phase: default, 20260508, 12345, 99999.

**Additional figures.**

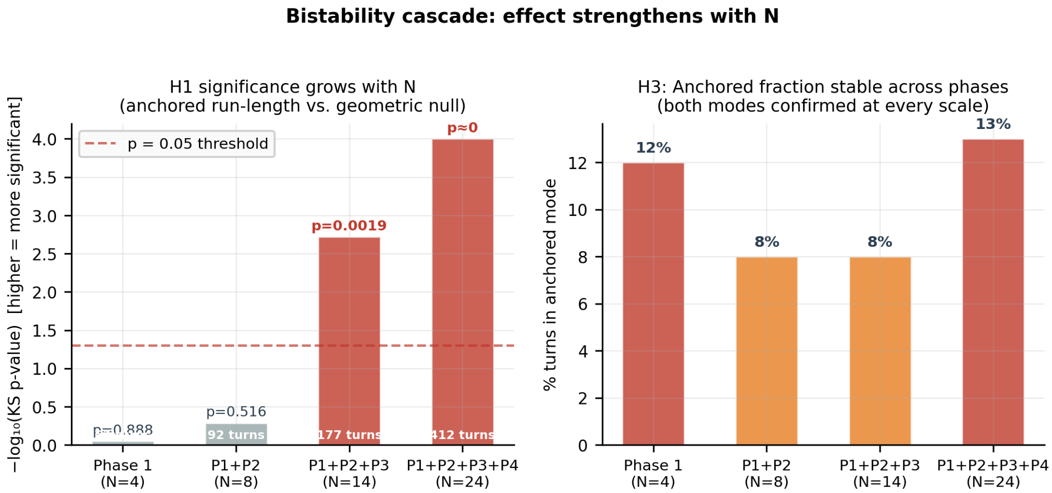


Figure 4: **Mode-switching evidence accumulates across phases.** *Left:* H1 significance (bootstrap p-value) grows with $N$ as more anchored runs become available to test against the geometric null. The power increase from $N = 14$ to $N = 24$ reflects both additional conversations *and* Phase 4's longer conversations (max_shards $= 10$), so this trajectory is not a fully independent replication. *Right:* Anchored fraction remains 8–14% at every scale, confirming both modes are persistently present.

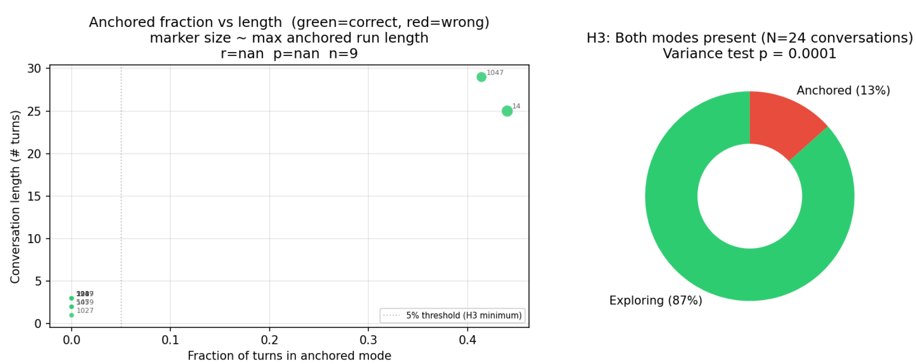


Figure 5: **Per-conversation summary.** *Left:* Anchored fraction vs. conversation length; marker size encodes the longest anchored run. Longer conversations tend to have higher anchored fractions (see also Section 4.9 for the length-stratified analysis). *Right:* Global H3 distribution (13% anchored / 87% exploring) with the H3 variance test p-value.

## Conference For AI Scientists 2026 — AI Involvement Checklist

### Research Stage Assessment

1. **Hypothesis development**

   Answer: **[D]**

   Iteration Effort: **[Medium]**

   Explanation: The research question (do CoT traces become unfaithful in multi-turn conversations?) was proposed by an AI research agent (Claude Sonnet) after surveying the CoT faithfulness and multi-turn conversation literatures. The human researcher selected this question from three candidates. The mode-switching reframing emerged from the agent's interpretation of preliminary data from sample 965.

2. **Experimental design and implementation**

   Answer: **[D]**

Iteration Effort: **[Medium]**

Explanation: The AI agent designed and implemented the full experimental pipeline: model wrapper, counterfactual deletion measurement, batch runner, and mode-switching analysis scripts. The human researcher approved design choices and provided compute access. Several iterations handled GPU memory constraints, Unicode issues, and metric bugs (including replacing an invalid KS test with a parametric bootstrap after reviewer feedback).

3. **Analysis of data and interpretation of results**

   Answer: **[C]**

   Iteration Effort: **[Low]**

   Explanation: The AI agent ran quantitative analyses (bootstrap tests, chi-square tests, confound checks) and generated all figures. Interpretation of the mode-switching pattern—connecting anchored turns to premature commitment—was developed jointly.

4. **Writing**

   Answer: **[D]**

   Iteration Effort: **[Low]**

   Explanation: The paper was written by the AI agent following the Research Paper Writing Skill guidelines (one message per paragraph, claim-evidence alignment, adversarial self-review). The human researcher reviewed for accuracy.

**AI System Documentation**

5. **AI systems used**

   Description: A frontier large language model acting as an autonomous research agent (Lossfunk Autoresearch harness, Claude Sonnet 4.x). The agent surveyed literature, proposed hypotheses, implemented experiments, ran them via SSH on a GPU server, interpreted results, and drafted this paper with high autonomy under human oversight.

6. **Observed AI Limitations**

   Description: (1) The agent initially designed experiments at a scale infeasible on a shared GPU, wasting hours before scope was reduced. (2) A metric bug (correctness-flip instead of answer-consistency) required a debugging pass. (3) The agent used 512 max_new_tokens for faithfulness regeneration (146 s/turn) before the human researcher identified that 128 was sufficient (31 s/turn, $5\times$ faster). (4) The agent's initial H1 test used scipy's `kstest` on discrete data, which produces a guaranteed-significant KS statistic due to a left-boundary artifact; this was replaced with a parametric bootstrap and chi-square GOF. All failures were caught through human review of intermediate outputs.

## Conference For AI Scientists 2026 — Reproducibility and Responsibility Checklist

1. **Claims**

   Answer: [Yes]

   Justification: Abstract and Introduction accurately reflect scope. H3 (13% anchored; variance test $\chi^2 = 58.5$, $p < 0.001$, $N = 24$) is confirmed; H1 (run-length persistence) is explicitly stated as inconclusive (bootstrap $p = 0.557$, $\chi^2$ $p = 0.218$, $N = 24$); H2 is stated as reserved for future work, not failed. All claims are supported by Section 4.

2. **Limitations**

   Answer: [Yes]

   Justification: Section 6 discusses single model, single task, context-solvability confound, H2 reserved for future work, and cross-model scale.

3. **Theory assumptions and proofs**

   Answer: [NA]

   Justification: No theoretical claims requiring proof.

4. **Experimental result reproducibility**

   Answer: [Yes]

   Justification: Appendix A specifies model versions, quantisation, seeds, data files, and hyperparameters.

5. **Open access to data and code**

   Answer: [Yes]

   Justification: Dataset from public `microsoft/lost-in-conversation` repository. Both models publicly available on HuggingFace. Code included in anonymised supplementary material.

6. **Experimental setting/details**

   Answer: [Yes]

   Justification: Appendix A provides all hyperparameters, infrastructure details, and selection criteria.

7. **Experiment statistical significance**

   Answer: [Yes]

   Justification: Section 4 reports bootstrap-corrected and chi-square H1 p-values, the H3 variance test, and explicitly characterises H2 as reserved for future work (not failed).

8. **Experiments compute resources**

   Answer: [Yes]

   Justification: Appendix A: NVIDIA RTX 6000 Ada (49 GB VRAM), 8/4-bit quantisation, $\approx$20 GPU-hours total.

9. **Research Ethics**

   Answer: [Yes]

   Justification: Publicly available models and datasets. No human subjects or sensitive data.

10. **Broader impacts**

    Answer: [Yes]

    Justification: Section 5 discusses the safety implication (CoT unreliable during anchored phases) and argues for greater caution when using CoT for oversight. No negative societal impacts specific to this work.